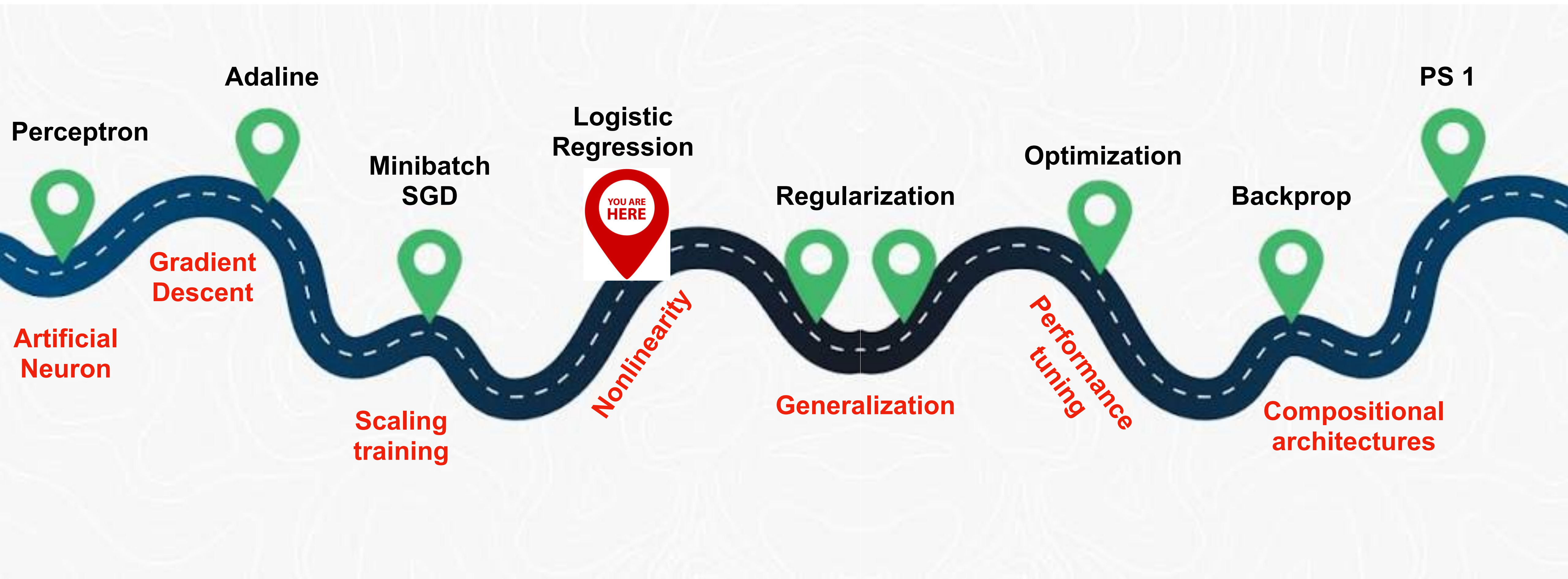




Logistic Regression & Regularization

Data 442 / 610 Lecture 4

Roadmap to Deep Learning



Outline

- Logistic regression and maximum likelihood estimation
- L1 and L2 Regularization
- *Maximum a posteriori* estimation

Relevant reading: *Deep Learning* Chapter 5, sections 2, 5, 6

Logistic Regression

- Adaline uses a continuous objective $\mathcal{L}$ and linear activation $\sigma(z) = z$
- Logistic regression uses a nonlinear activation to predict class probabilities
- The **preactivation** is a logit (log-odds)

$$z \equiv \text{logit}(p) = \log \frac{p}{1-p}$$

Preactivation: $z = \mathbf{w}^\top \mathbf{x} + b$

Sigmoid: $\sigma(z) = \frac{1}{1 + e^{-z}}$

$$\sigma(z) : (-\infty, \infty) \rightarrow (0, 1)$$

Logistic Regression: Loss function

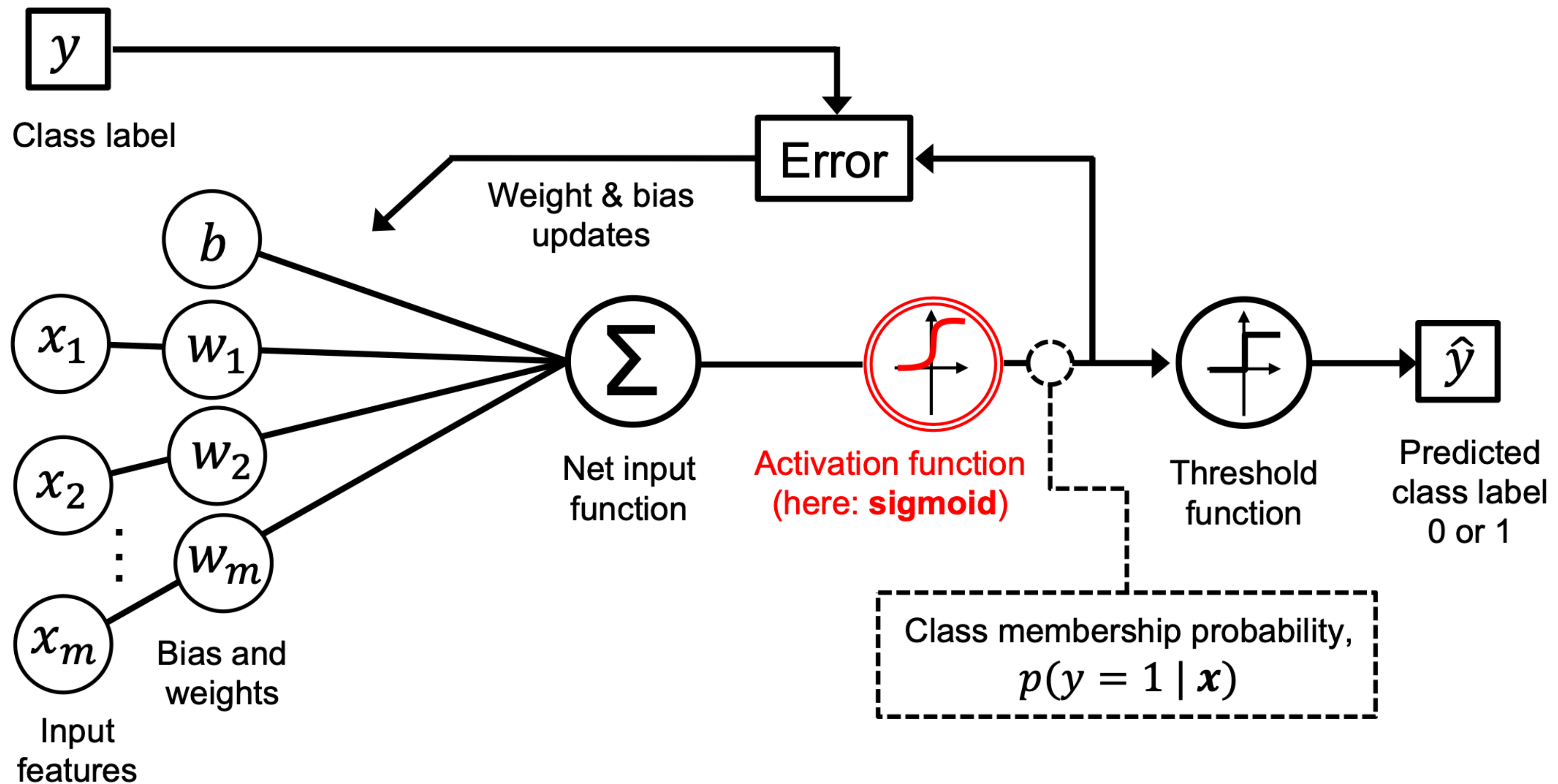
Goal: Find $\mathbf{w}$ to maximize the *likelihood* of observations $\mathbf{w}^* = \operatorname{argmax}_{\mathbf{w}} \left[p(y | \mathbf{x}; \mathbf{w}, b) \right]$ **Maximum Likelihood Estimation**

$$p(y | \mathbf{x}; \mathbf{w}, b) \sim \prod_i p(y^i | \mathbf{x}^i; \mathbf{w}, b) = \prod_i \sigma(z^i)^{y^i} \cdot (1 - \sigma(z^i))^{1-y^i} \quad \text{Bernoulli distribution}$$

$$\mathcal{L} = \sum_i -y^i \log \sigma(z^i) - (1 - y^i) \log (1 - \sigma(z^i)) \quad \text{Cross entropy loss}$$

Goal: Find $\mathbf{w}$ to minimize the *cross entropy* loss $\mathcal{L} = \sum_i -y^i \log \sigma(z^i) - (1 - y^i) \log (1 - \sigma(z^i))$

Logistic Regression



Courtesy of C. Fanelli

Hands-on Demo: Logistic Regression



Multiclass Regression: Softmax

- Logistic regression extends to multiple classes

$$z_j = \sum_k w_{jk} \cdot x_k + b_k$$

- Each preactivation is an un-normalized log-probability

$$z_j \equiv \log \tilde{p}$$

- The corresponding activation is the **softmax** function

$$\text{softmax } z_j = \frac{\exp z_j}{\sum_k \exp z_k}$$

Numerically-stable softmax

$$\text{softmax } (z_j - \max \mathbf{z}) = \text{softmax } z_j$$

Views z_j as a log-odds with respect to the most-probable outcome

$$z_j \sim \log \frac{p_j}{p_{\max}}$$

Linear Regression: Loss function

Goal: Find $\mathbf{w}$ to maximize the *likelihood* of observations $\theta^* = \operatorname{argmax}_{\theta} p_{\theta}(y | \mathbf{x})$ **Maximum Likelihood Estimation**

$$p_{\theta}(y | \mathbf{x}) \sim \prod_i p_{\theta}(y^i | \mathbf{x}^i) = \prod_i \mathcal{N}(y^i; \hat{y}_{\theta}^i, \sigma^2)$$

Gaussian distribution

$$\mathcal{L} = \frac{1}{N} \sum_i \|y^i - \hat{y}^i\|^2$$

Mean Squared Error Loss

Goal: Find $\mathbf{w}$ to minimize the *mean squared error* loss $\mathcal{L} = \frac{1}{N} \sum_i \|y^i - \hat{y}^i\|^2$

More about MLE

What we have: $p_{\text{data}}(\mathbf{x}, y)$

What we want: $p_{\theta}(y | \mathbf{x})$

Kullback-Leibler Divergence

$$D_{KL}(p(x) \parallel q(x)) = \mathbb{E}_{x \sim p(x)} \left[\log \frac{p(x)}{q(x)} \right] \geq 0$$

$$\begin{aligned} \text{Minimize } D_{KL}(p_{\text{data}}(x, y) \parallel p_{\theta}(y | x) p_{\text{data}}(x)) &= \mathbb{E} \left[\log \frac{p_{\text{data}}(x, y)}{p_{\theta}(y | x) p_{\text{data}}(x)} \right] && \geq 0 \\ &= \mathbb{E} [\log p_{\text{data}}(x, y) - \log p_{\theta}(y | x) - \log p_{\text{data}}(x)] && \geq 0 \\ &= \mathbb{E} [\log p_{\text{data}}(y | x) - \log p_{\theta}(y | x)] && \geq 0 \end{aligned}$$

Model negative log-likelihood

MLE is our “best guess” at the underlying data distribution!

Of course it's not so simple

- No Free Lunch Theorem (Wolpert, 1996): “any two optimization algorithms are equivalent when their performance is averaged across all possible problems”
- We don't want the model with the best training error, we want the model that generalizes best to new data
- Thus, we have to make a determination about *what our problem is* and *how to modify our algorithm* to best generalize for that problem

Regularization is any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error

Regularization

L1 Normalization (LASSO): $\mathcal{L}_{L1}(\mathbf{w}) = \mathcal{L}(\mathbf{w}) + \alpha \sum_j |w_j|$

L2 Normalization (Ridge): $\mathcal{L}_{L2}(\mathbf{w}) = \mathcal{L}(\mathbf{w}) + \frac{\alpha}{2} \sum_j w_j^2$

ElasticNet: $\mathcal{L}_{EN}(\mathbf{w}) = \mathcal{L}(\mathbf{w}) + \alpha \left(\beta \sum_j |w_j| + (1 - \beta) \sum_j w_j^2 \right)$

$$L_p\text{-norm} : \|\mathbf{z}\|_p := \left[\sum_j |z_j|^p \right]^{1/p}$$

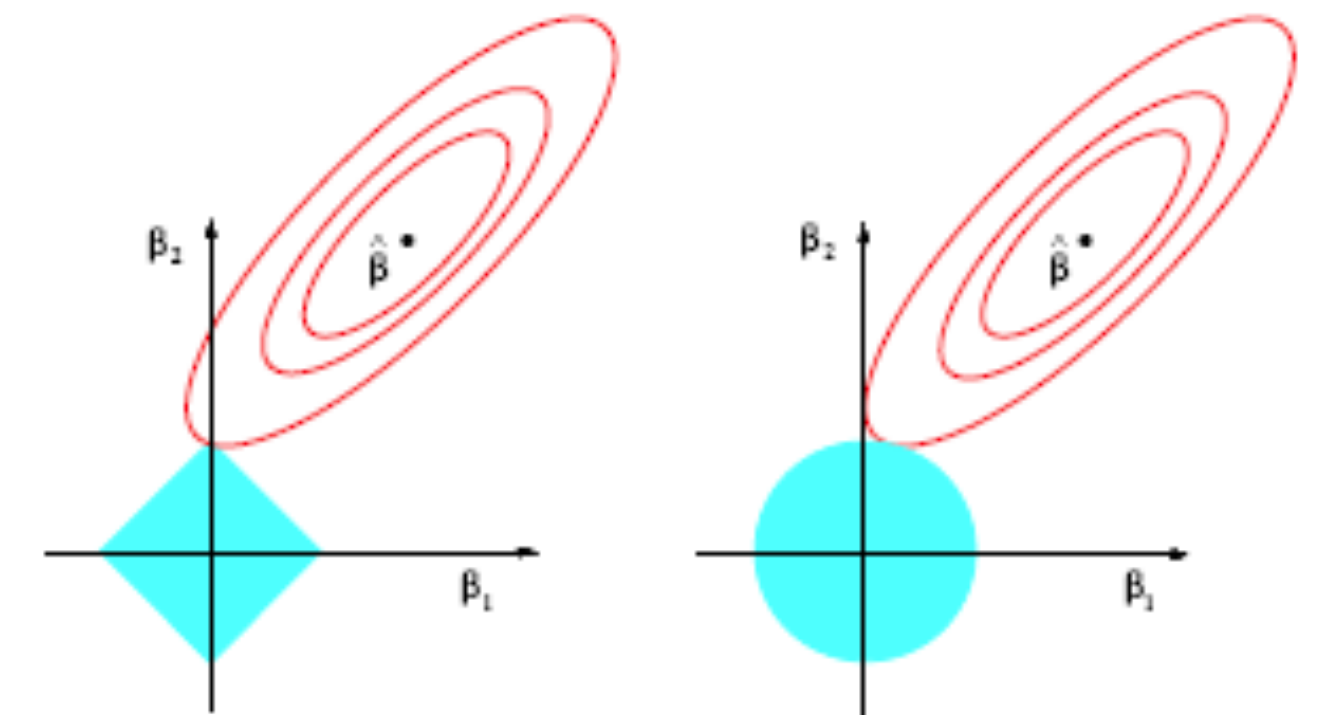


Figure 3.12: Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq t$ and $\beta_1^2 + \beta_2^2 \leq t^2$, respectively, while the red ellipses are the contours of the least squares error function.

Work out how this affects weight updates on the whiteboard

Hands-on Demo: Regularization



That's all!

- Slides will be posted on the course website: <https://colen-teaching.github.io/DATA442-Fall-2026>
- Lab demos will also be posted
- **PS0** due this Friday, September 11
- Look out for **PS1** later this week!